

I am in the process of exploring Unsupervised Machine Learning

I completed Supervised Machine Learning Algorithms
Now I am exploring in Unsupervised Machine Learning Algorithm

I have covered clustering algorithms

Today I will be starting with Dimensionality Reduction
Unsupervised Machine Learning Algorithms

What is Principal Component Analysis and first of all what is dimensionality reduction?

Explain in detail from Zero to very advanced. Make sure the explanation should be in pointers only, use very easy going and simple language such that even 15 year old student can understand. Make use of example for better understanding. Starting form the perspective of 15 year old student take the explanation to ML engineer level. Go from explanation to mathematical implementation then coding in python.



Great progression 

I'll take you **from absolute zero** → **ML engineer level**, **only in pointers**, **simple language**, **lots of examples**, and then **math + Python code**.

1. First: What is Dimensionality? (Zero Level – 15-year-old view)

- Dimension = number of features (columns) in data
- Example:
 - A student record:
 - Height
 - Weight

- Age
 - Marks
 - 👉 This data has **4 dimensions**
 - Another example (images):
 - A **28 × 28 image** = 784 pixels
 - Each pixel = one feature
 - 👉 Image has **784 dimensions**
 - **More features = higher dimensional data**
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2. What is Dimensionality Reduction? (Beginner Level)

- Dimensionality Reduction = reducing number of features
- Goal:
 - Keep **important information**
 - Remove **unimportant or repeated information**

Simple analogy (Very important 👉)

- Imagine:
 - You wrote **10 pages of notes**
 - Teacher says: "Make a **2-page summary**"
 - You:
 - Remove repeated points
 - Keep main ideas
 - 👉 That is **Dimensionality Reduction**
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3. Why Do We Need Dimensionality Reduction?

Problems with high-dimensional data:

- Data becomes **slow**
- Needs **more memory**
- Models **overfit easily**
- Visualization becomes impossible

- Distance-based models behave badly
(called **Curse of Dimensionality**)
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4. Real-Life Example (Very Easy)

- Suppose you want to predict **fitness**
 - Features:
 - Height
 - Weight
 - Waist size
 - BMI
 - BMI is already made using:
 - Height + Weight
 - 👉 Information is **repeated**
 - Dimensionality reduction removes this repetition
-

5. Types of Dimensionality Reduction

- Feature Selection
 - Remove features manually or using rules
 - Feature Extraction ✅ (PCA belongs here)
 - Create **new features**
 - Old features are combined
-

6. Now: What is PCA? (Principal Component Analysis)

One-line definition:

- PCA is an unsupervised algorithm that reduces dimensions by finding new directions where data varies the most
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7. PCA Explained for a 15-Year-Old

- Imagine:
 - You throw coins on the floor
 - Coins form a **long stretched shape**
 - Instead of:
 - Measuring X and Y separately
 - You:
 - Measure **along the longest direction**
 - 👉 That direction captures **most information**
-

8. Key Idea of PCA (Core Concept)

- PCA finds:
 - **New axes (directions)**
 - Where data variance is maximum
 - These new axes are called:
 - **Principal Components**
-

9. What Are Principal Components?

- Principal Component = **new feature**
 - Properties:
 - Always **perpendicular** to each other
 - Ordered by importance:
 - PC1 → most information
 - PC2 → second most
 - PC3 → less
 - ...
-

10. Visual Example (2D → 1D)

- Original features:
 - X-axis = Height
 - Y-axis = Weight

- PCA creates:
 - PC1 = diagonal direction
 - We keep only PC1
 - 👉 Data becomes 1D instead of 2D
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11. What PCA Actually Does (Simple Steps)

- Step 1: Mean center the data
 - Step 2: Find directions of maximum variance
 - Step 3: Sort directions by importance
 - Step 4: Keep top K directions
 - Step 5: Project data onto them
-

12. Why PCA Is Unsupervised?

- PCA:
 - Does NOT use labels (y)
 - Only looks at X
 - Hence:
 - Unsupervised Machine Learning
-

13. Benefits of PCA

- Faster training
 - Less memory
 - Removes noise
 - Better visualization
 - Helps models perform better
-

14. Limitations of PCA

- Hard to interpret new features
- Assumes:

- Linear relationships
 - Sensitive to scale
 - Loses some information
-

15. Transition to Math (Beginner → Intermediate)

Key idea in math:

- PCA tries to **maximize variance**
 - Variance = how spread out data is
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16. Mathematical Foundation (Intermediate Level)

Step 1: Mean Centering

- Formula:
 - $X_{centered} = X - \mu$
 - Why?
 - PCA works around origin (0)
-

Step 2: Covariance Matrix

- Covariance tells:
 - How two features change together
- Formula:

$$\text{Cov}(X) = \frac{1}{n-1} X^T X$$

Step 3: Eigenvalues & Eigenvectors

- Eigenvectors → **directions**
- Eigenvalues → **importance of direction**
- PCA:
 - Sort eigenvectors by eigenvalues (descending)

Step 4: Select Top K Components

- Keep top K eigenvectors
 - These form:
 - **Projection matrix**
-

Step 5: Projection

$$Z = X_{centered} \cdot W$$

- Z = reduced data
 - W = selected eigenvectors
-

17. Explained Variance Ratio (Advanced Concept)

- Tells:
 - How much information is kept
- Formula:

$$\frac{\lambda_i}{\sum \lambda}$$

- Example:
 - PC1 = 70%
 - PC2 = 25%
 - PC3 = 5%
-

18. Choosing Number of Components

- Methods:
 - Explained variance $\geq 95\%$
 - Scree plot
 - Elbow method
-

19. PCA vs Feature Selection

- PCA:
 - Creates new features
 - Feature selection:
 - Keeps old features
-

20. PCA in Machine Learning Pipelines (Engineer View)

- Common usage:
 - Image compression
 - NLP embeddings
 - Anomaly detection
 - Preprocessing before clustering
 - Visualization (2D / 3D)
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21. Important Rule Before Using PCA

- Always scale data
 - Use:
 - StandardScaler
 - Reason:
 - PCA is variance-based
-

22. Python Implementation (Beginner Friendly)

python

 Copy code

```
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.datasets import load_iris
```

```
# Load data
```

```
X, y = load_iris(return_X_y=True)
```



```
# Scale data
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Apply PCA
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

print(X_pca.shape)
```

23. Explained Variance in Python

python

 Copy code

```
print(pca.explained_variance_ratio_)
```

24. Visualizing PCA

python

 Copy code

```
import matplotlib.pyplot as plt

plt.scatter(X_pca[:, 0], X_pca[:, 1], c=y)
plt.xlabel("PC1")
plt.ylabel("PC2")
plt.show()
```

25. PCA from Scratch (Advanced Engineer Level)

python

 Copy code

```
import numpy as np

# Mean center
X_meaned = X - np.mean(X, axis=0)
```

```
# Covariance matrix
cov_matrix = np.cov(X_meaned, rowvar=False)

# Eigen decomposition
eigen_values, eigen_vectors = np.linalg.eigh(cov_matrix)

# Sort
sorted_index = np.argsort(eigen_values)[::-1]
eigen_vectors = eigen_vectors[:, sorted_index]

# Select top k
k = 2
W = eigen_vectors[:, :k]

# Project data
X_reduced = X_meaned.dot(W)
```

26. When NOT to Use PCA

- When features are already meaningful
 - When interpretability is critical
 - When data is non-linear (use t-SNE, UMAP)
-

27. PCA vs t-SNE vs UMAP (Quick)

- PCA → Linear, fast, interpretable
 - t-SNE → Visualization only
 - UMAP → Better structure preservation
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28. Final Intuition Summary (15-Year-Old Style)

- PCA:
 - Finds smarter directions
 - Removes repeated info

- Makes data smaller & faster
 - Think of it as:
 - **Smart compression for data**
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If you want, next we can go **deep** into:

- PCA geometric intuition with diagrams
- Scree plot interpretation
- PCA + Clustering together
- Comparison with Autoencoders
- Interview-level PCA questions

Just tell me 🚀



Answer in very short

Distance-based models behave badly
(called Curse of Dimensionality)

Means Unsupervised Clustering ML Algos?



Yes — mainly **Unsupervised distance-based algorithms**.

Very short explanation:

- **Distance-based models** = models that rely on distance calculations
- **Mostly affects Unsupervised Clustering algorithms**, such as:
 - K-Means
 - Hierarchical Clustering
 - DBSCAN
 - Mean Shift

Why?

- In high dimensions:

- Distances between points become almost equal
- "Nearest" and "farthest" points look the same
- Clusters become unclear

Also affects (but not only unsupervised):

- KNN (supervised)
- Any model using Euclidean / cosine distance

👉 So yes, mainly **Unsupervised Clustering ML algorithms**, but not limited to them.



In very very short tell me what is variance



Variance = how spread out the data is from its average (mean).



